$= ON - OM$ (Measuring the distance from O)

$= x_2 - x_1$ ……..(1)

And $RQ = NQ - NR$

$= NQ - MP$ (Opposite sides of the rectangle $MNRP$)

$= y_2 - y_1$ ……..(2)

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Step 2

Triangle PQR is right angled at R. ($PR \perp NQ$)

$PQ^2 = PR^2 + RQ^2$ (By Pythagoras theorem)

$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

$d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$ (Taking positive square root)

Note

Distance between two points

- Given two points $P(x_1, y_1)$ and $Q(x_2, y_2)$, the distance between these points is given by the formula $d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$.
- The distance between PQ = The distance between QP
  i.e. $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2} = \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$
- The distance of a point $P(x_1, y_1)$ from the origin $O(0,0)$ is $OP = \sqrt{x_1^2 + y_1^2}$

5.3.4 Properties of Distances

We have already seen that distance (A,B) = distance (B,A) for any points A, B on the plane. What other properties have you noticed? In case you have missed them, here are some:

distance $(A,B) = 0$ exactly when A and B denote the identical point: $A = B$.

distance $(A,B) > 0$ for any two distinct points A and B.

Now consider three points A, B and C. If we are given their co-ordinates and we find that their x-co-ordinates are the same then we know that they are collinear, and lie on a line parallel to the y-axis. Similarly, if their y-co-ordinates are the same then we know that they are collinear, and lie on a line parallel to the x-axis. But these are not the only conditions. Points (0,0), (1,1) and (2,2) are collinear as well. Can you think of what relationship should exist between these coordinates for the points to be collinear?

The distance formula comes to our help here. We know that when A, B and C are the vertices of a triangle, we get,

distance(A,B) + distance(B,C) >distance(A,C) (after renaming the vertices suitably).

When do three points on the plane not form a triangle? When they are collinear, of course. In fact, we can show that when,

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